BIOCHEMISTRY CUMULATIVE EXAM

April 3, 1993

Cytochrome c is a highly basic protein containing 19 lysines and 2 arginines. It is very soluble in water and completely insoluble in hydrocarbons such as hexane. However, if cytochrome c is added to an aqueous solution containing cardiolipin lipid bilayers in 10 mM phosphate buffer, pH 7, and mixed in a separatory funnel containing hexane, a very unusual thing happens. An appreciable amount of cytochrome c is extracted into the hexane layer as monitored by the red color. For every molecule of cytochrome c extracted, about 10 to 12 molecules of cardiolipin are also extracted. This extraction does not occur if phosphatidyl choline lipid bilayers are used instead of cardiolipin bilayers.

A. (40) Propose a detailed model for the structure of the cytochrome c - cardiolipin adduct extracted into hexane.

B. (30) Propose a series of detailed experiments that would prove or disprove your model for the adduct.

C. (30) Cytochrome c is impermeable to lipid bilayers composed of neutral lipids. However, it has been found to be transported across bilayers containing cardiolipin at a slow by measurable rate. Propose a mechanism for how this transport might occur.